

The State of On-Site Construction Risk: Why 50% of Field Incidents Go Unmitigated

A deep-dive analytical report examining safety blind spots, data fragmentation across subcontractors, and how automated capture frameworks reduce critical liability on active job sites.

1 in 5	\$1B+	65%	2–4x
workplace fatalities occur in construction	paid weekly in workers' comp (US)	of fatalities from the "Fatal Four"	indirect costs exceed direct costs

Key Findings

This report identifies three structural failures in how construction firms manage field-level risk — failures that allow preventable incidents to escalate into costly liabilities.

Finding 1: The documentation gap is systemic, not incidental

Industry data suggests that a significant share of field incidents — from near-misses to minor injuries — are never formally documented. Without structured documentation, teams cannot conduct root cause analysis, identify recurring patterns, or demonstrate regulatory compliance.

Finding 2: Subcontractor data silos amplify risk invisibly

On multi-employer job sites, each subcontractor generates its own safety data — if it generates any at all. This fragmentation means the general contractor lacks a unified view of site risk.

"It can cost you bids, it can cost you opportunities, and it can create rework challenges. But it goes further than that — safety, project costs, pre-construction. Each one presents its own challenges in data alignment."

— Brad Buckles, VP Technology & Innovation, Charles Perry Partners

Finding 3: The cost multiplier effect is real — and underestimated

OSHA estimates indirect costs — lost productivity, crew retraining, project delays — reach 2–4x the direct cost. Liberty Mutual found employers paid \$1B+/week in direct workers' comp alone.

"Workers' comp gets filed and everyone moves on. What doesn't get counted is the superintendent who spent two days dealing with it instead of running the job."

Analysis: The Fragmentation Problem

A single project can involve an owner, architect, engineers, a GC and a dozen subcontractors — each running their own systems. Every handoff is a chance for data to break.

- **Visibility failure:** Leadership cannot see full site risk when data is scattered across WhatsApp, paper logs, and spreadsheets.
- **Analysis failure:** Without centralized data, trend analysis and safety benchmarking across projects is impossible.
- **Compliance failure:** Fragmented records force expensive timeline reconstruction during audits and litigation.

Automated Capture Framework

Organizations transitioning from manual to automated capture report measurable improvements:

	Manual Capture	Automated Capture
Time to document	15–30 min per incident	< 60 seconds (photo + AI)
Data completeness	40–60% of fields filled	95%+ via AI extraction
Audit readiness	Reconstructed retroactively	Real-time, always current
Pattern detection	Manual, quarterly at best	Automated, continuous
Institutional memory	Lost between projects	Cumulative knowledge base

Recommended Actions

- 1 Mandate structured digital capture**
Replace paper logs and WhatsApp with a standardized digital workflow. Must work in under 60 seconds or field adoption fails.
- 2 Centralize subcontractor safety data**
Require all subs to report through a single platform. Fragmented data creates invisible risk accumulation.
- 3 Automate pattern detection**
Use AI analytics to surface recurring categories, high-risk phases, and trades with disproportionate events.

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Measure documentation ROI

Track time saved, incidents prevented, compliance gaps closed. The business case is quantifiable within 90 days.

BaceTech Perspective

BaceTech addresses these findings through a unified capture-to-insight platform for active job sites. Teams capture incidents via photo or voice in seconds; AI generates structured descriptions, identifies causes, recommends solutions, and references applicable standards. All data centralizes across projects and time — building institutional knowledge that makes each project safer. Available for 30-day free evaluation at **app.bace-tech.fr**

Sources: OSHA Safety Pays Framework; Liberty Mutual 2025 WSI; BLS CFOI 2023; Autodesk Construction Disconnected Report; OpenSpace 2026; Achilles 2025.

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